higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

MARKING GUIDELINE

NATIONAL CERTIFICATE (VOCATIONAL)

CONCRETE STRUCTURES
NQF LEVEL 2

12 MARCH 2019

This marking guideline consists of 6 pages.

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QUESTION 1: GENERAL INTRODUCTION TO FORMWORK AND FALSEWORK

1.1 True
1.2 False
1.3 True
1.4 True
1.5 False
1.6 True
1.7 False
1.8 False
1.9 False
1.10 False

(10 × 1) **[10]**

QUESTION 2: FORM FACE MATERIALS AND SPECIAL TYPES OF FORMS

2.1 2.1.1
- Steel is very versatile – thousands of uses are possible.
- Steel leaves a very smooth concrete surface.
- Steel is extremely strong.
- Steel can be used as permanent formwork that is left in place.

(4 x 1) (4)

2.1.2
- Steel is expensive.
- The thicker the surface of the steel sheet the greater the weight.
- Must use a releasing agent.
- Steel dents relatively easy. (4 x 1) (4)

2.2
- Permeable material absorbs fluids.
- Non-permeable material does not absorb fluids easily. (2)

2.3 2.3.1 Multiple-storey buildings (1)

2.3.2
- Lifted by elevator
- Flown by crane (2 x 1) (2)

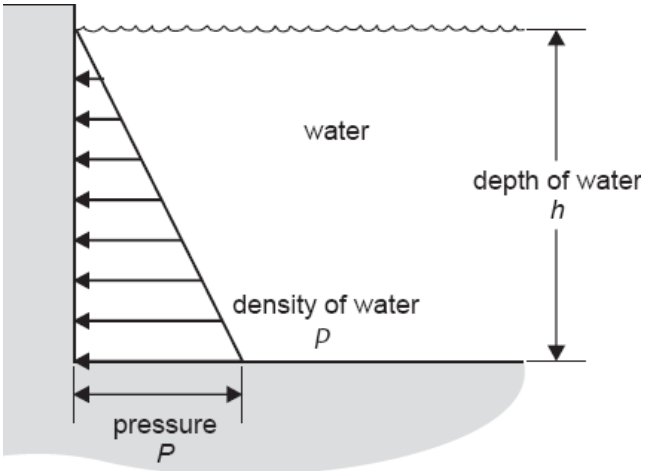
2.4
- Permanent formwork can be manufactured off-site.
- Some types of permanent formwork can be set up quickly.
- Using permanent formwork does away with the need to take apart formwork and false work.
- Using permanent formwork improves safety by reducing hazards during construction.
- Using permanent formwork reduces construction waste produced during construction. (Any 2 × 1) (2)

[15]

QUESTION 3: PRESSURES RESISTED BY FORMWORK AND FALSEWORK

3.1 Pressure = Density × Gravity × Hight
= 1 000kg/m^3 × 9,8 m/s^2 × 8 m✓✓✓
= 78 480 N/m^{-2} or 78,48 kNm^{-2}✓✓
Or = 78 480 Pa or 78,48 kPa (5)

3.2



(4)

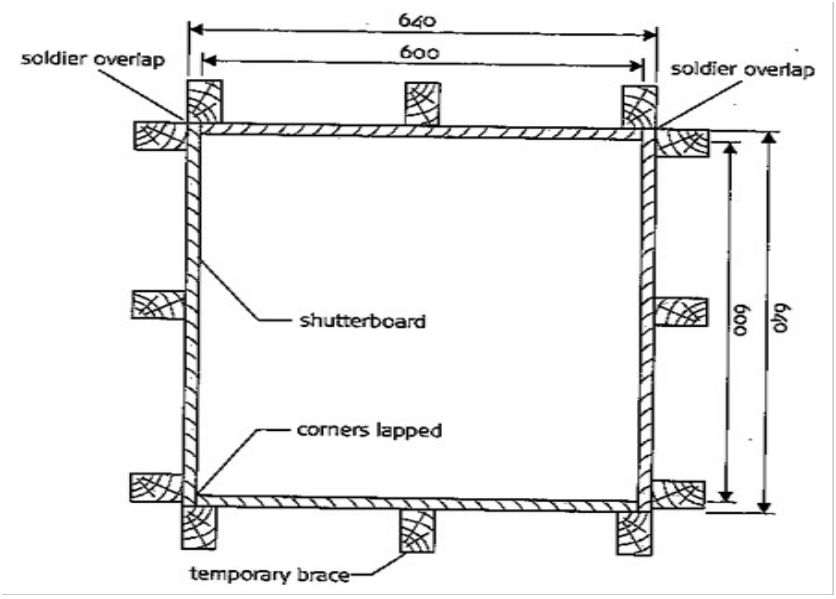
3.3 Snap ties are removed by breaking them off. (1)
[10]

QUESTION 4: ERECTING FORMWORK AND FALSEWORK

4.1
4.1.1 Plate
4.1.2 Prop inner
4.1.3 Hole for course adjustment
4.1.4 Tube
4.1.5 Prop pin with retainer
4.1.6 Fin adjustment collar
4.1.7 Cut thread
4.1.8 Prop outer (base)

(8 × 1) (8)

4.2



Correctness (3)
Labelling (6)
Measurements (2)

4.3
- Position the sole plates according to the building plan and the engineer's specifications.
- Then erect the support work for the slab formwork in the same way as you erected the beam support work.
- Check that all support work is level, stable and doesn't wobble.
- Clean and tidy your work area. (4 x 1) (4)

4.4
- Check the building plan and engineer's specifications.
- Make sure that you have all tools, equipment and material at hand.

(2 x 1) (2)
[25]

QUESTION 5: VERTICAL STOP ENDS, WATER STOPS AND JOINTS

5.1 5.1.1 Used where a fresh pour of concrete is next to or touches an existing one. They are stopping places in the process of construction, because a construction joint is the stopping of work for the day.

5.1.2 Allows relative movement of structure parts that are next to each other in a controlled or uncontrolled way without affecting the quality or working of the structure

5.1.3 Used to separate new concrete from existing or adjacent construction

(3 × 2) (6)

5.2
- Prepare well in advance
- Pour straight through until the job is finished (2 x 1) (2)

5.3
- Flat wheel
- Plant failure
- Bad weather (Any relevant 2 × 1) (2)

[10]

QUESTION 6: USING RELEASING AGENTS

6.1
- Avoid contact with skin.
- Avoid breathing.
- Don't get in contact with food.
- Wear protective clothing.
- Follow the manufacturer's instructions.

(Any other reasonable answer 4 × 1) (4)

6.2
- Rug
- Paintbrush
- Roller
- Sprayer (4)

6.3 24 hours (1)

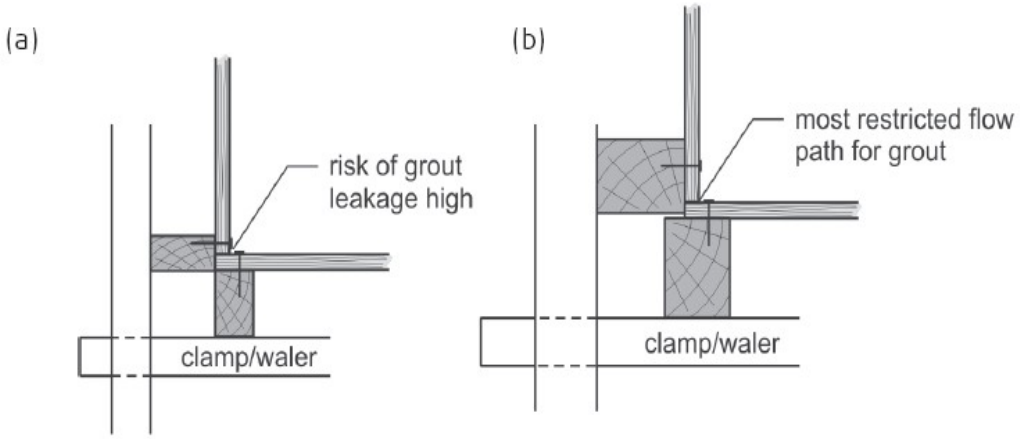
6.4 28 days (1)

[10]

QUESTION 7: WORKMANSHIP AND DETAILING IN CONCRETE STRUCTURES

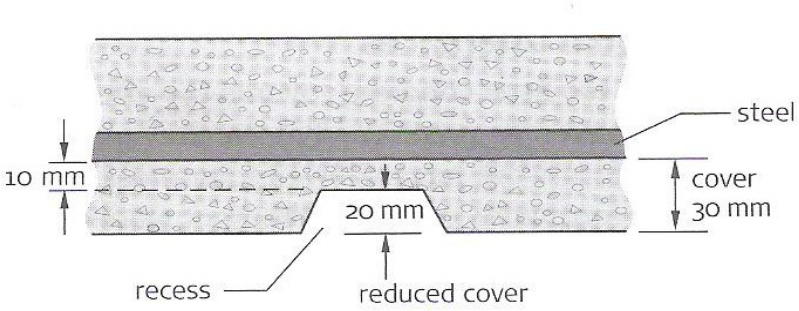
7.1
- Geometric dimension of the formwork
- Accuracy of the horizontal and vertical surfaces
- All formwork must be clean (from dust, mud, dirt, nails, et cetera)
- All joints must be flush, tight and sealed
- No nails sticking out
- Water stops must be correct
- Correct number of ties
- Tie lengths must be correct
- Ties must be tight enough
- All releasing agents must be applied correctly. (Any 6 × 1) (6)

7.2



Drawing correct (6)
Labelling (4)

7.3



(4)
[20]

TOTAL: 100

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